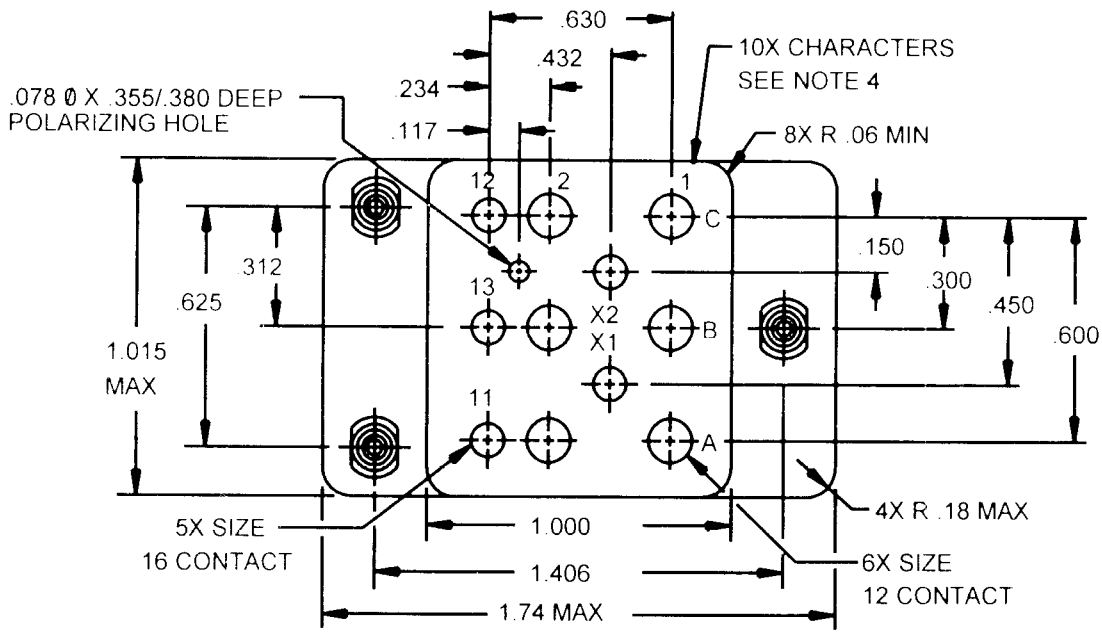
PERFORMANCE SPECIFICATION SHEET

SOCKETS, PLUG-IN ELECTRONIC COMPONENTS
SOCKET FOR MIL-R-6106 RELAYS AND MS27742 RELAYS

This specification is approved for use by all Departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-12883.

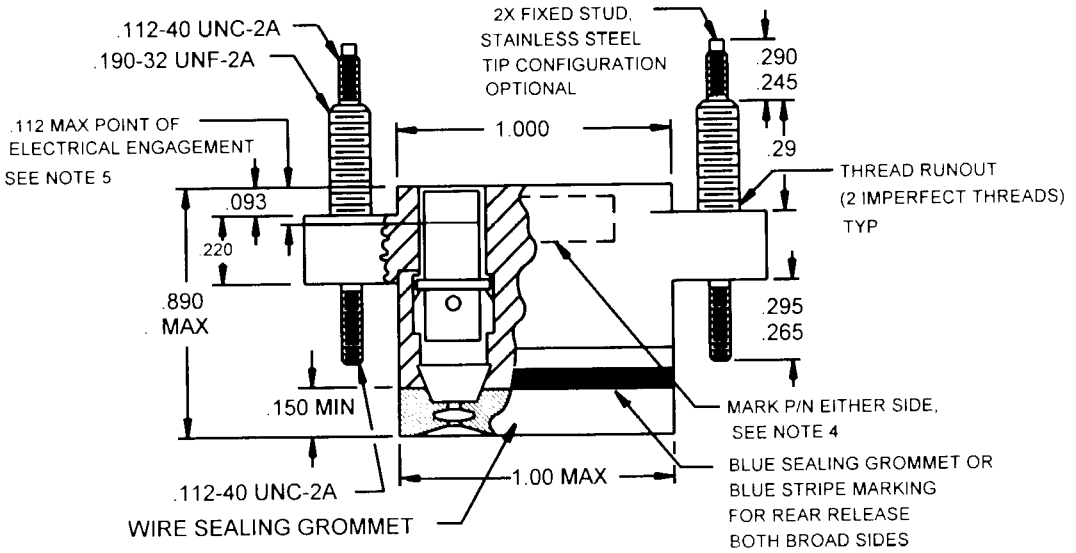


Mating face

FIGURE 1. Socket configurations (-01)

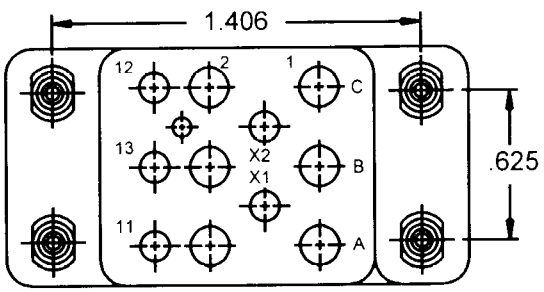
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Front view

-01 through -04



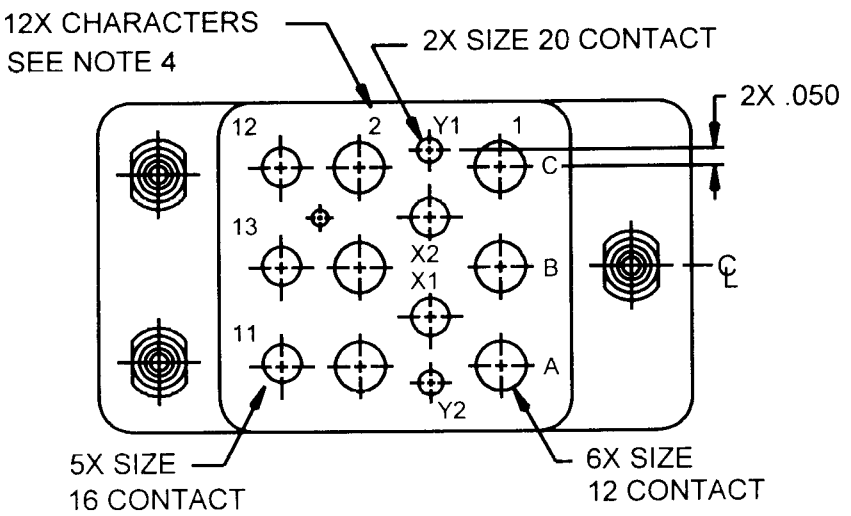
Mating face

-02

All other dimensions same as -01

FIGURE 1. Socket configurations - Continued.

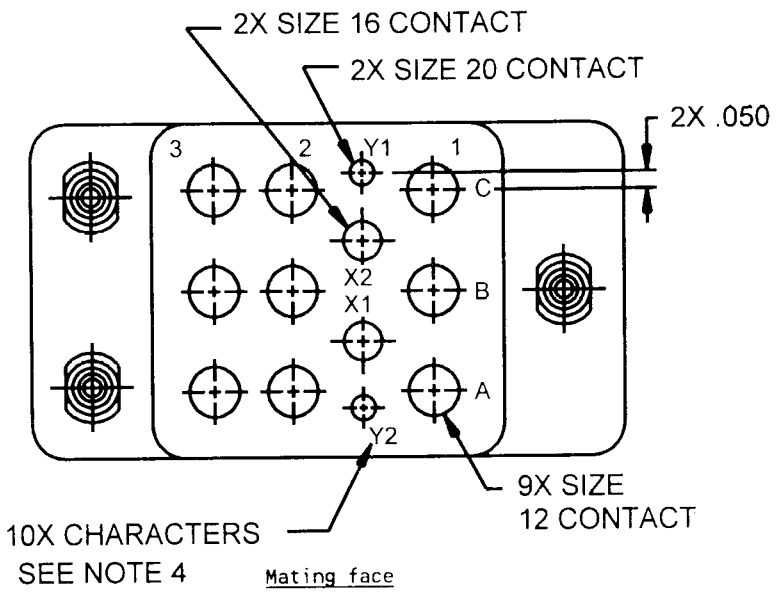
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Mating face

-03

All other dimensions same as -01



Mating face

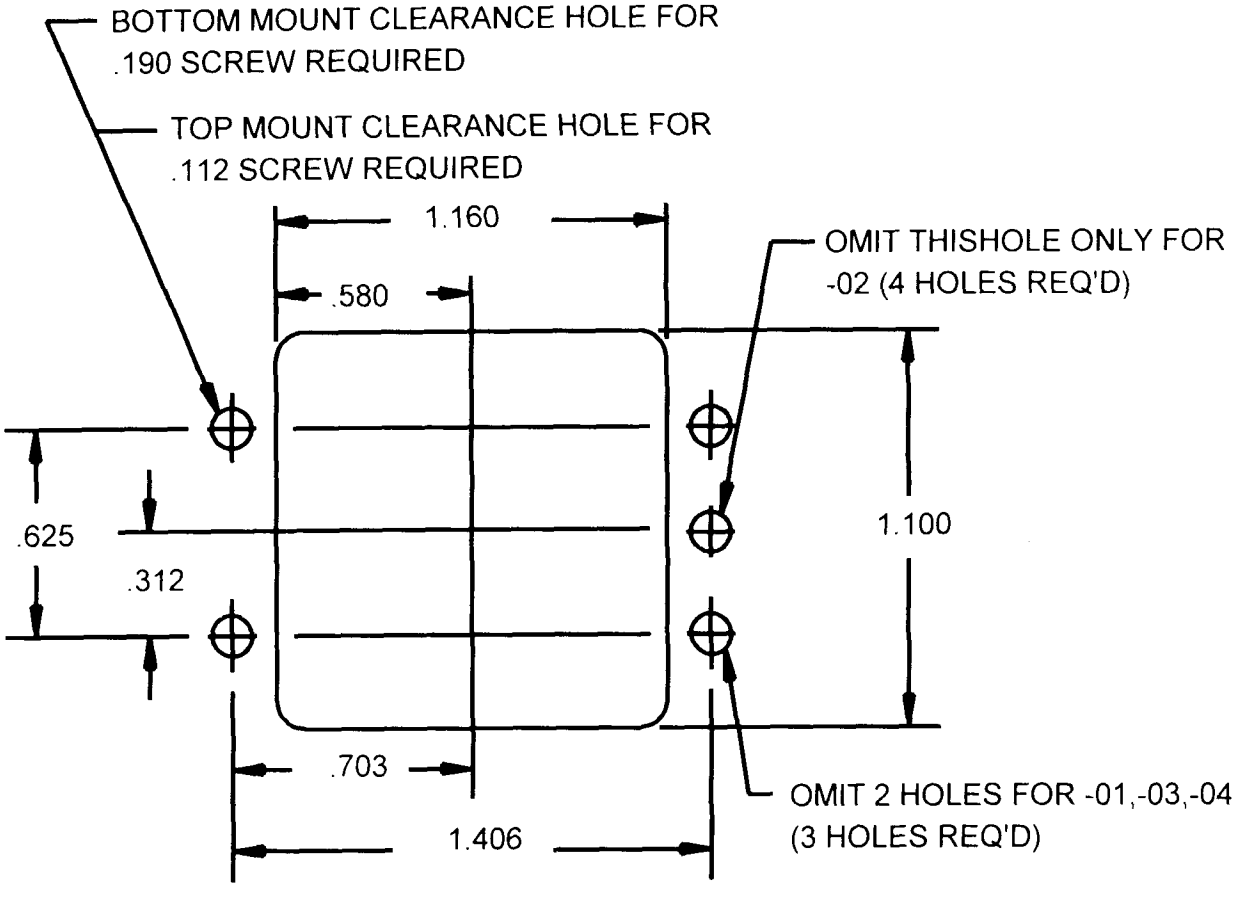
-04

All other dimensions same as -01

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.01 (0.25 mm) for two place decimals.
4. Marking shall be characters which are molded .035 (0.89 mm) minimum, ink marking optional; see MIL-STD-1285.
5. Point or electrical contact from mating face of socket insulator to the socket contact.
6. Configuration mounting (see figure 2).

FIGURE 1. Socket configurations - Continued.

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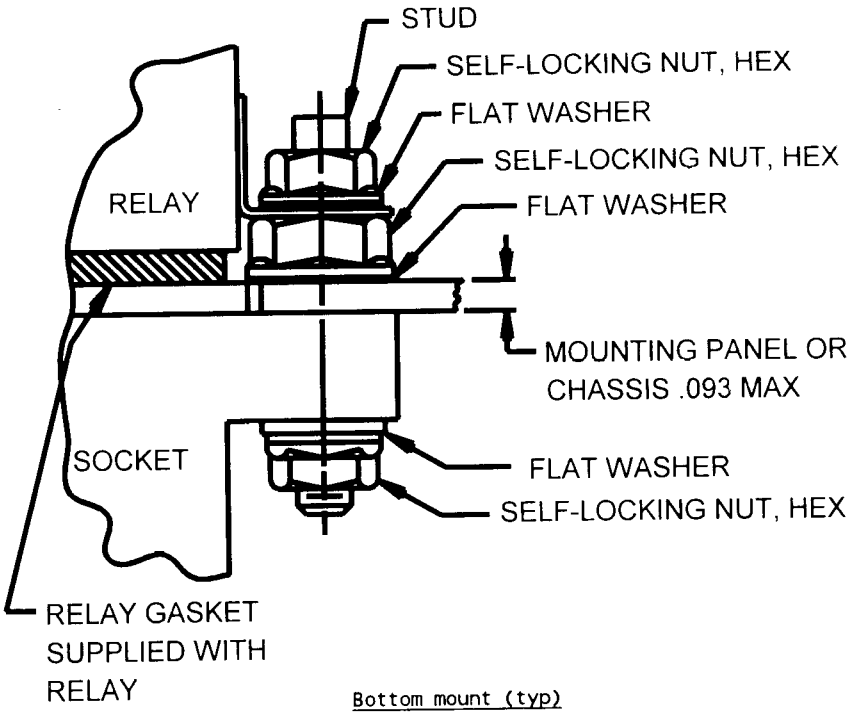


Recommended mounting dimensions

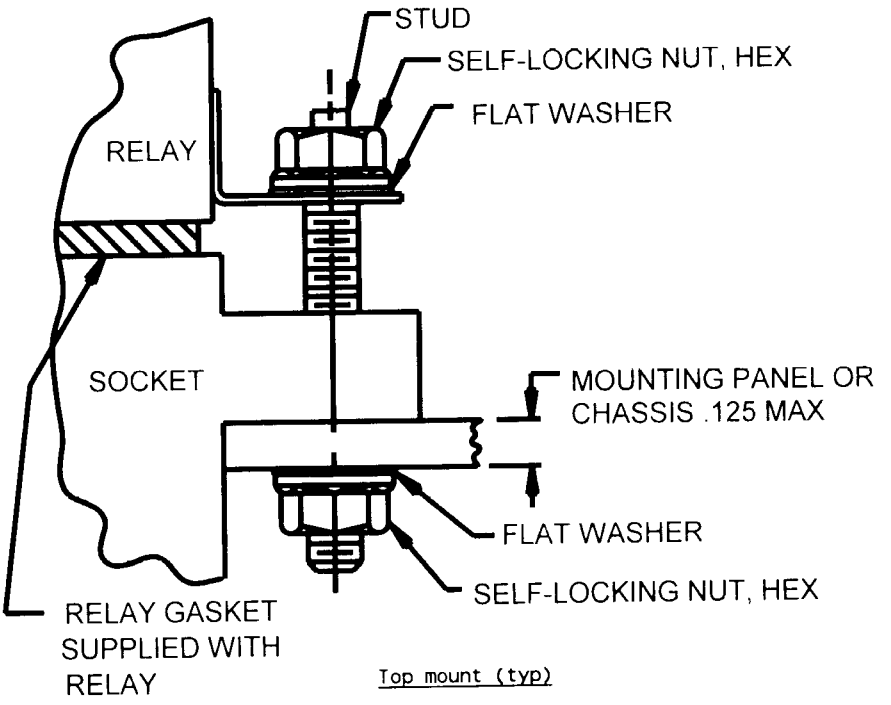
Inches	mm	Inches	mm	Inches	mm
.005	0.13	.150	3.81	.890	22.61
.01	0.25	.220	5.59	1.000	25.40
.03	0.8	.234	5.94	1.015	25.78
.035	0.90	.300	7.62	1.025	26.04
.06	1.5	.312	7.92	1.085	27.56
.093	2.36	.432	10.97	1.250	31.75
.112	2.84	.450	11.43	1.406	35.71
.135	3.43	.600	15.24	1.71	43.4
.140	3.56	.625	15.88	1.718	43.64
.15	3.8	.630	16.00		

FIGURE 2. Socket mounting.

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Bottom mount (typ)



Top mount (typ)

FIGURE 2. Socket mounting - continued.

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REQUIREMENTS:

Design and construction: See figures 1, and 2.

Reference material: (see 3.1.1 of MIL-PRF-12883)

Insulator: Diallyl phthalate in accordance with MIL-M-14, type SDG-F or any glass filled thermoplastic material in accordance with MIL-P-46179.

Color: Color shall be optional providing that the color provides a contrasting background for the blue sealing grommet/blue color bands indicating rear release contacts.

Grommet: Silicone rubber in accordance with ZZ-R-765.

Mounting hardware: Corrosion resistant steel or steel with cadmium/chromate finish.

Electrical:

Contacts: Contacts shall be removable crimp type in accordance with MIL-C-39029/92, or approved equivalent contacts manufactured to a commonly accepted industry standard (see table I).

Insulation resistance: 1000 megohms minimum. Test pin diameter, size 12, .0940 ±0.0010 inch (2.39 ±0.025 mm); test pin diameter, size 16, .0620 ±.0010 inch (1.575 ±0.025 mm); size 20, .0400 ±.0010 inch (1.02 ±0.03 mm).

TABLE I. Dash numbers and configuration.

Dash Number	Contact size		Number of Contacts	Contact Designation M39029	Mating relay
	Mating end	Wire barrel			
01	16 12	16 12	5 6	/92-533 /92-535	M6106/13-002, -008, -011, -013, -015
02	16 12	16 16	5 6	/92-533 /92/536	M6106/13-004
03	20 16 12	20 16 12	2 5 6	/92-532 /92-533 /92-535	M6106/12-002
04	20 16 12	20 16 12	2 2 9	/92-532 /92-533 /92-535	MS27742-2

Dielectric withstanding voltage:

Sea level: Test voltage, 1500 V rms.

High altitude (80, 000 feet (24,384 meters)): Test voltage, 500 V rms.

Test pin diameter, size 20, .0400 ±.0010 inch (1.02 ±0.03 mm), size 16, 0.0625 ±0.0010 inch (1.595 ±0.025 mm), size 12, .0940 ±.0010 inch (2.39 ±0.025 mm).

Environmental:

Temperature range: Operating temperature range -70°C to +125°C.

Wire sealing: A resilient grommet is permanently bonded to the wire entry face of the socket so as to provide moisture sealing capabilities for AWG size 20, 16, and 12 wire as applicable (see figure 1, front view).

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Mechanical:

Vibration sinusoidal): In accordance with MIL-STD-202, method 204, test condition G, or an equivalent industry standard.

a. Except that the frequency range shall be varied logarithmically between the limits of 10 Hz and 3,000 Hz.

b. Except that the procedure of method 201 of MIL-STD-202 may be applied during 10 Hz to 55 Hz band of the vibration frequency range.

c. Qualified mating relay used as a test gauge.

Vibration (random): In accordance with MIL-STD-1344, method 2005, test condition V, letter G, or an equivalent industry standard, with a test duration 15 minutes. The mating relay shall be used as the test gage.

Shock (mechanical): In accordance with MIL-STD-202, method 213, condition C, or an equivalent industry standard, except peak value shall be 200g's.

Mounting hardware: The mounting hardware shall allow mounting the socket above or below the panel or chassis (see figure 2), and shall allow mounting and securing the relay to the socket without disturbing the mounted socket or access to the wiring side of the socket. The hardware shall provide the nominal spacing between socket surface and relay mounting flange, regardless of mounting configuration.

Supplied with relay socket:

2 each per mounting stud;

.112-40 self locking nuts (.206 max dia x .176 max height).

.112 flat washers (.220 max O.D. x .018 max thick).

1 each per mounting stud;

.190-32 self locking nuts (.330 max dia x .190 max height).

.190 flat washers (.360 max O.D. x.019 max thick).

Fixed mounting studs: Studs shall be fixed into the mounting flange of the socket and shall be designed so as to prevent rotation of the stud within the flange (see figure 2).

Insertion and withdrawal forces: The insertion and withdrawal forces of the relay and socket shall be as specified in table II.

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TABLE II. Insertion and withdrawal forces.

M12883/54	-01, -02	-03	-04
Initial Insertion force (max)	21 lbf (93.41 N)	23 lbf (102.31 N)	25 lbf (111.21)
Withdrawal force (min)	2.0 lbf (8.90 N)	2 lbf (8.90 N)	2 lbf (8.90 N)
After 10 insertions and withdrawals, before vibration Insertion force (max)	23 lbf (102.31 N)	25 lbf (111.21 N)	27 lbf (120.10 N)
Withdrawal force (min)	2.0 lbf (8.90 N)	2 lbf (8.90 N)	2 lbf (8.90 N)
After vibration Insertion force (max)	23 lbf (102.31 N)	25 lbf (111.21 N)	27 lbf (120.10 N)
Withdrawal force (min)	2.0 lbf (8.90 N)	2.0 lbf (8.90 N)	2 lbf (8.90 N)

Contact installation tools: see table III, approved equivalent industry standard tools may also be used where appropriate.

TABLE III. Contact installation tools.

Nomenclature	Part or Identifying Number (PIN) (size 20-20 contacts)	PIN (size 16-16) contacts	PIN (size 12-12, 12-16) contacts
Crimp tool Positioner	M22520/2-01 M22520/2-02	M22520/1-01 M22520/1-02	M22520/1-01 M22520/1-02
Insertion/removal tool Unwired Wired	M81969/30-05 M81969/14-02	M81969/30-06 M81969/14-03	M81969/30-07 M81969/14-04

Torque: Relay socket and hardware shall be subjected to testing as specified in table IV. Sockets shall be installed in mounting panel when test torque is applied. No physical damage shall be permitted. Torque shall be maintained for a reasonable period of time to insure stud, socket, and associated hardware have not been damaged (see table IV).

TABLE IV. Torque requirements (installed in panel conditions).

Thread size	Torque			
	Testing		Installation	
	Inch-pounds	Newton-meters	Inch-pounds	Newton-meters
.112-40	8 + 1 - 0	0.90 + 0.11 - 0	4 ±1	0.45 ±0.11
.190-32	24 + 1 - 0	2.71 +0.11 - 0	18 ±1	2.03 ±0.11

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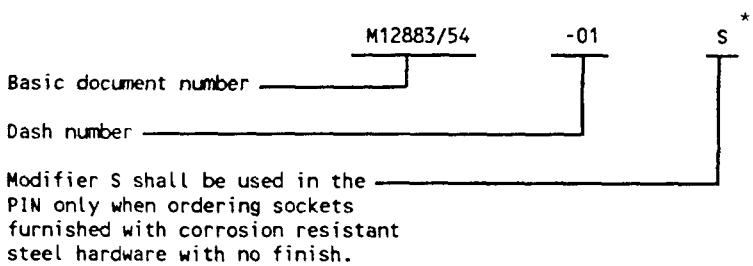
Weight: Maximum weight of relay, socket, all contacts and all associated hardware shall be as specified in table V.

TABLE V. Weight.

Part designations M12883/54	Maximum weight	
	Pounds	Grams
-01 and -02	.132	60
-03 and -04	.154	70

PIN: The PIN shall be marked on the socket body as shown in the example (see figure 1). The PIN shall consist of the basic number of this specification sheet and the dash number from table I.

Example



* For acquisition of these sockets, parts identified with an "S: modifier shall be corrosion resisting steel (CRS), and parts without an "S" modifier shall be cadmium chromate finish. No mixing of hardware types shall be permitted.

Ordering data: Sockets without contacts may be ordered when so indicated in the ordering data (see MIL-PRF-12883). This applies only to original equipment manufacturers (OEM's) and subcontractors. All direct shipments to the government shall include all applicable contacts and mounting hardware. The PIN to be marked on the socket shall be as shown in the PIN example (see figure 1 and table I).

The Government PIN, specified in table VI, supersedes the following commercial part numbers.

TABLE VI. Supersession and cross reference.

Active Government PIN	Superseded number
	CAGE 58982
M12883/54-01	RSE116627
M12883/54-02	RSE116631
M12883/54-03	RSE116629
M12883/54-04	RSE116635

CONCLUDING MATERIAL

Custodian:
Navy - EC
ARMY - CR
Air Force - 85

Review activity:

Preparing activity:
DLA - ES

(Project 5935-4035-01)

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